

Test Yourself

Microeconomics

Equilibrium/Disequilibrium

Section A

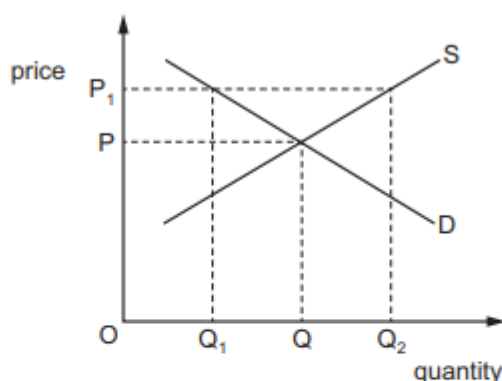
Duration of each question is **40 mins**

1. With the help of a diagram, assess whether the benefits of a buffer stock scheme for an agricultural product always outweigh its disadvantages. (12) **q3b/21/M/J/23**
2. Assess the effectiveness of government intervention through buffer stock schemes in agricultural markets. (12) **q3b/23/M/J/23**
3. With the help of a diagram, assess whether the introduction of a minimum price in a market can be justified. (12) **q3b/22/F/M/23**
4. Governments sometimes fix maximum prices on goods for different reasons. Assess whether the use of a maximum price is always advantageous. (12) **q3b/21/O/N/23**
5. A government wishes to keep the price of an essential food, such as rice, affordable to help low-income households. Assess whether a policy of fixing the maximum price for an essential food is likely to be more effective than a policy of making transfer payments to low-income households. (12) **q2b/22/M/J/23**

Section B

Duration 1 hour

1. The diagram shows the demand and supply curves for a good.

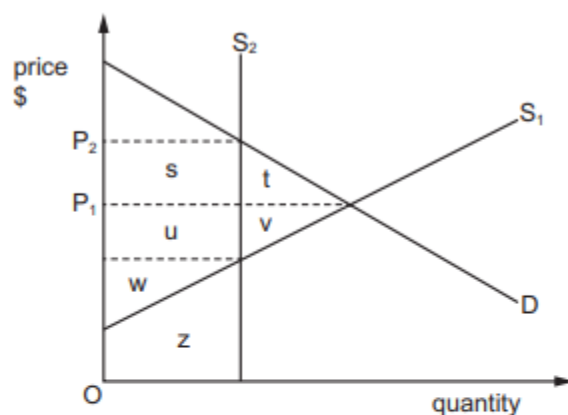


The government fixes a maximum price of OP_1 .

What would happen?

- A** Consumers would have to be rationed to quantity OQ_1 .
- B** The government would have to introduce a subsidy of PP_1 .
- C** The market equilibrium quantity OQ would be demanded and supplied.
- D** The supply of quantity OQ_2 would be guaranteed.

2. The diagram shows a shift of the supply curve in a market from S_1 to S_2 .

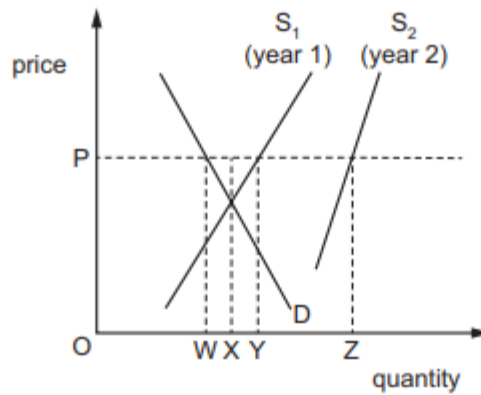


How will the consumer surplus and producer surplus change?

	consumer surplus	producer surplus
A	decreases by s	increases by $s + u - t$
B	decreases by s	increases by $u + v - w$
C	decreases by $s + t$	increases by $s + t - v$
D	decreases by $s + t$	increases by $s + z - v$

3. In the diagram, S_1 and S_2 are the supply curves for an agricultural product in years 1 and 2 respectively. D is the demand curve in years 1 and 2.

In year 1, the government purchased an amount necessary to ensure that the price was OP .

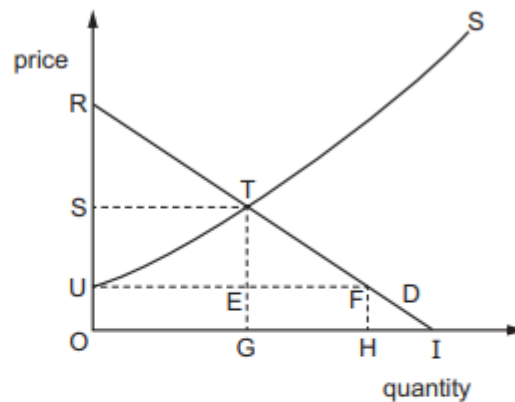


The price is held at OP in year 2.

How much more must the government buy in year 2 than it bought in year 1?

- A** WZ **B** XY **C** XZ **D** YZ

4. Different areas on the diagram represent different aspects of a market.



Which row correctly identifies the three variables shown?

	consumer surplus	producer surplus	total consumer expenditure
A	RST	TUOG	STEU
B	RST	UST	STGO
C	RTU	UTE	STGO
D	RTEU	TEF	RFU

5. A free market is in disequilibrium with a shortage of a product.

As the market moves towards equilibrium, what will happen to the price, the quantity demanded and the quantity supplied?

	price	quantity demanded	quantity supplied
A	decrease	decrease	increase
B	decrease	increase	decrease
C	increase	decrease	increase
D	increase	increase	decrease

6. Which statement correctly describes the area of producer surplus on a demand and supply diagram?

- A** It is above the supply curve and below the equilibrium price line.
- B** It is above the supply curve and below the demand curve.
- C** It is below the demand curve and above the equilibrium price line.
- D** It is below the supply curve and above the horizontal axis.

7. In a free market, there is a surplus of a good.

Which change would cause the market to come to an equilibrium?

- A** a decrease in demand
- B** a fall in price
- C** a government minimum price
- D** an increase in supply

8. Which event is most likely to enable firms to respond to an increase in demand by increasing supply rather than by raising price?

- A** Firms reduce their expenditure on the training of workers.
- B** Import duties are increased on machines that increase the speed of production.
- C** The government introduces buffer stock schemes for the goods produced by the firm.
- D** Trade unions become strong, leading to poor labour productivity.

9. The equation for the quantity demanded, Q_D , for a product is

$$Q_D = 400 - 20P$$

where P = price in \$. The quantity supplied, Q_S , is given by $Q_S = 100 + 40P$.

Which change will occur if the price rises from \$5 to \$6?

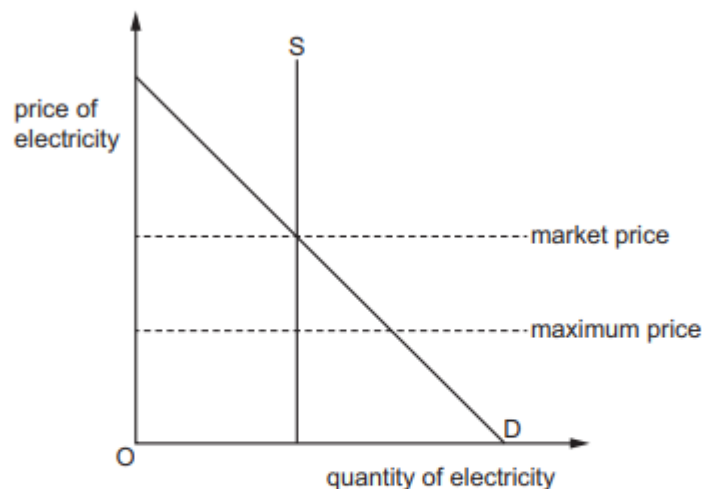
- A The market will move from equilibrium to shortage.
 - B The market will move from equilibrium to surplus.
 - C The market will move from shortage to surplus.
 - D The market will move from surplus to shortage.
10. A free market is currently experiencing excess supply at a price of P_1 .
- What will be the effect on the current market situation if a maximum price is set above P_1 ?
- A The excess supply will increase.
 - B The market will move into an equilibrium position.
 - C There will be no effect.
 - D There will be excess demand.
11. What will increase the producer surplus of farmers that grow carrots?
- A a decrease in the demand for carrots
 - B a decrease in the price of carrot seed
 - C a decrease in the price of cabbage
 - D a decrease in the subsidy on carrots
12. The price of a product is above the market equilibrium price.
- Which combination of changes is certain to result as the market adjusts towards equilibrium?

	quantity demanded	quantity supplied
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

13. Which row shows the effect of the maximum and minimum prices described in the headings of the table?

	maximum price above equilibrium price	maximum price below equilibrium price	minimum price above equilibrium price
A	excess demand	no effect	no effect
B	excess supply	no effect	no effect
C	no effect	excess demand	excess supply
D	no effect	excess supply	excess demand

14. The diagram shows the market for electricity in a country that has a fixed supply of electricity and introduces a maximum price to make it affordable for poorer households.



What will be the effect of this?

- A** It will encourage producers to build more power stations in the future.
 - B** It will encourage the development of renewable sources of electricity.
 - C** It will increase producer surplus.
 - D** It will increase the probability of power cuts.
15. Which statement about market disequilibrium is correct?
- A** Price is likely to change.
 - B** Supply is equal to demand.
 - C** The government must intervene.
 - D** There must be shortages.

16. The demand and supply functions for a product are quantity demanded = $1500 + 50P$ and quantity supplied = $300P - 2000$ where P = price. With government regulation, the current price in the market is \$15.

What can be concluded about the form of price regulation and the balance of demand and supply in the market?

	form of price regulation	balance of demand and supply
A	effective maximum price	excess demand
B	effective maximum price	excess supply
C	effective minimum price	excess demand
D	effective minimum price	excess supply

17. A government imposes a maximum price on medical face masks, below the market equilibrium.

What is the most likely effect of this?

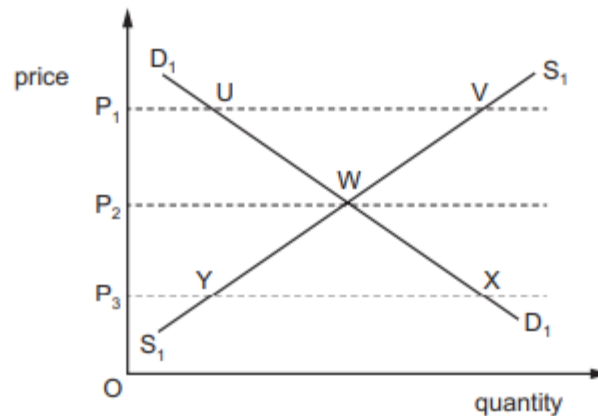
- A** Some consumers will be unable to buy face masks at the maximum price.
- B** Firms will produce more masks to meet the excess demand at the maximum price.
- C** Poorer consumers will be able to obtain all the masks they need at the maximum price.
- D** The market will clear at the maximum price.

18. A concert arena has a fixed capacity that allows it to sell 10 000 tickets for a concert. The current equilibrium price for a ticket is \$10. The owners of the arena decide to set a minimum price for a ticket.

Under which conditions will there be the most unsold tickets?

	minimum ticket price (\$)	price elasticity of demand for tickets
A	11	inelastic
B	11	elastic
C	12	inelastic
D	12	elastic

19. The diagram shows the market demand for and supply of a good.



Which statement is **not** valid?

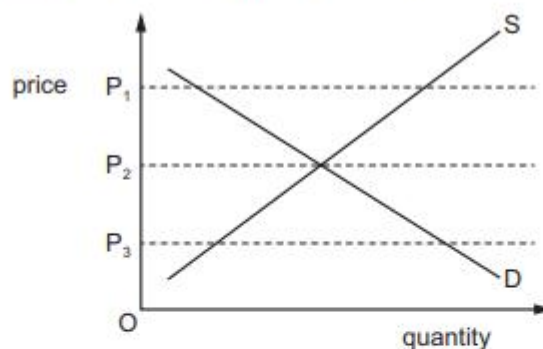
- A At price OP_1 , UV represents the market surplus.
 - B At price OP_3 , P_3X represents the quantity that consumers are able and willing to buy.
 - C If price were to fall from OP_1 to OP_3 , the extra quantity demanded would be equal to the extra quantity supplied.
 - D Price OP_2 is the market equilibrium price.
20. A government fixes a maximum price for a product in order to increase its consumption.
- What would be the likely outcome of such a policy?
- A Consumption will fall if the maximum price is above the current equilibrium price.
 - B Consumption will rise if the maximum price is below the current equilibrium price.
 - C Production will fall if the maximum price is above the current equilibrium price.
 - D Production will fall if the maximum price is below the current equilibrium price.
21. A market is in equilibrium with 100 units of the product sold at a price of US\$10 each. The price elasticity of supply for the product is +2.0 and the price elasticity of demand is -1.0.
- What will be the state of the market if a minimum price of US\$11 is imposed?
- A an excess demand of 10 units
 - B an excess demand of 30 units
 - C an excess supply of 20 units
 - D an excess supply of 30 units

22. A private company achieves a producer surplus at the market equilibrium price.

What might reduce the size of the private company's producer surplus?

- A Private companies choose to prioritise profits over customer needs.
- B Private companies start to pay dividends out of profits.
- C The government fixes the price of electricity below the market price.
- D The private company supply curve becomes more elastic.

23. The diagram shows a market that is currently in equilibrium.

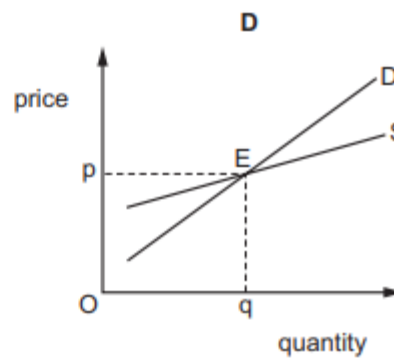
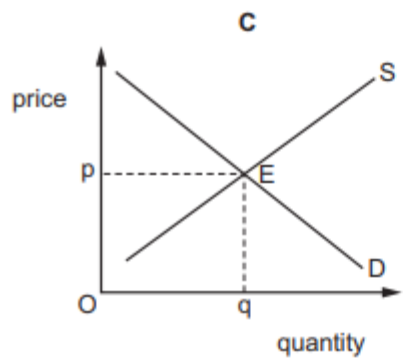
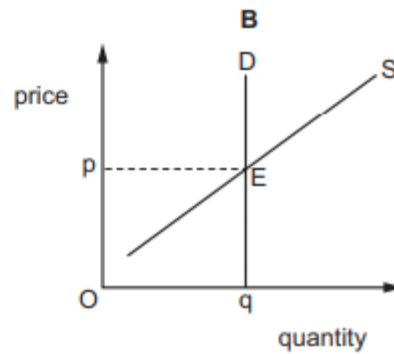
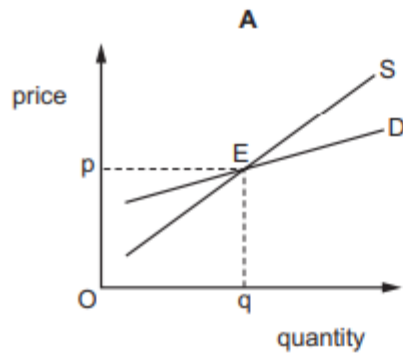


A government is considering the possible consequences of introducing a price control into this market.

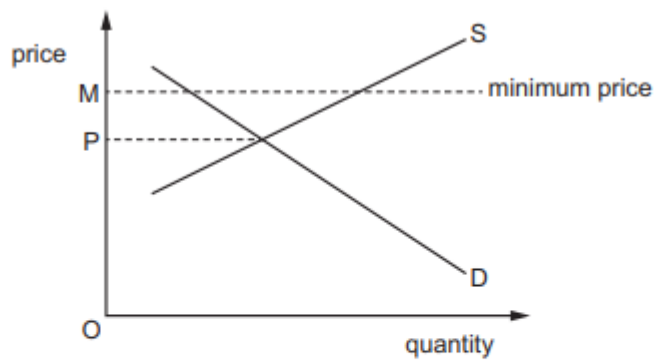
Which statement is **not** correct?

- A All consumers and some producers will lose if OP_1 becomes the minimum price.
 - B All consumers will be unaffected if OP_1 becomes the maximum price.
 - C No producers will gain if OP_2 becomes the maximum price.
 - D All producers will gain if OP_3 becomes the maximum price.
24. What would be the main effect of imposing an effective maximum price on foodstuffs?
- A Inflation would rise.
 - B The price of foodstuffs would rise.
 - C There would be a decrease in the import of foodstuffs.
 - D There would be a shortage of foodstuffs.

25. In which market would equilibrium **not** be achieved at point E if price were above O_p ?



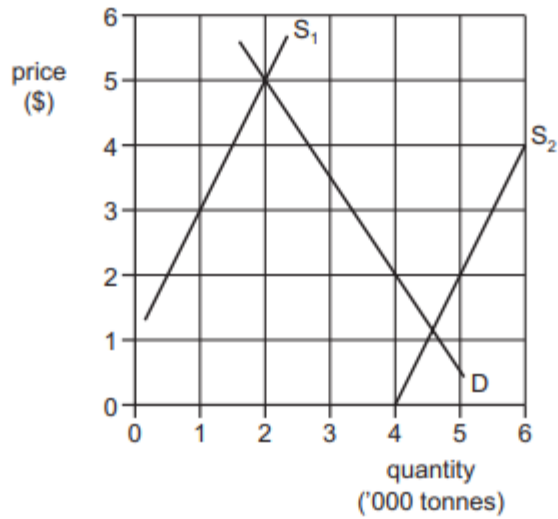
26. The diagram shows a market where a government has set a minimum price of OM .



What can be concluded from the diagram?

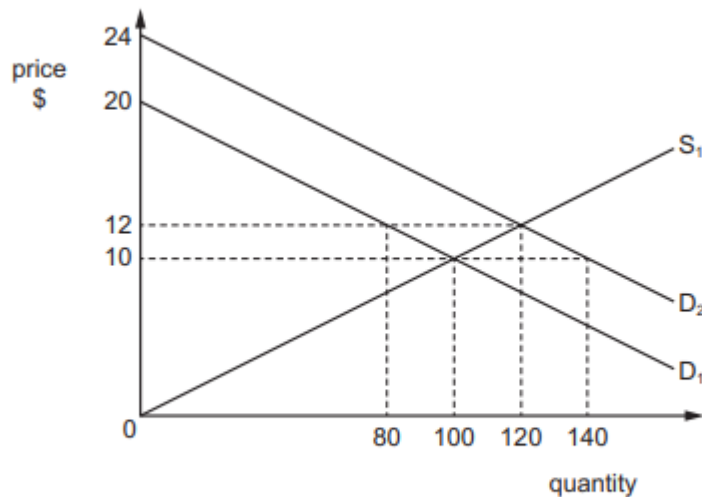
- A** At prices above OM , there would be a surplus of the good.
- B** At the minimum price OM , there would be a shortage of the good.
- C** Price will fall automatically from OM to OP .
- D** The minimum price will be ineffective.

27. In the diagram, D is the demand curve for an agricultural commodity and S_1 is the initial supply curve. A good harvest causes the supply curve to shift to S_2 .



By how much will the demand curve have to shift to leave farm incomes unchanged?

- A 500 tonnes at all prices
 - B 1000 tonnes at all prices
 - C 2000 tonnes at all prices
 - D 4000 tonnes at all prices
28. The diagram shows the market for a good with an initial equilibrium price of \$10. The demand for the good increases by 40 units at every price causing the equilibrium price to rise to \$12.



What is the value of the producer surplus after the price increase?

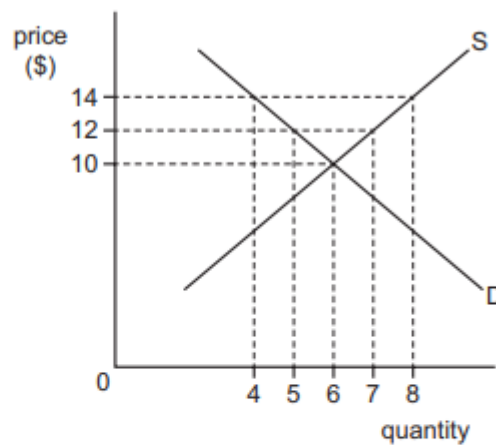
- A \$500
- B \$720
- C \$1000
- D \$1440

29. The quantity demanded of a product is given by $Q_D = 400 - 10P$, when P is the price in dollars. Supply of the product is fixed at 100 units.

If the price is \$20, what will be the position in the market?

- A It will be in disequilibrium with excess demand of 100 units.
- B It will be in disequilibrium with excess supply of 100 units.
- C It will be in equilibrium with 100 units traded.
- D It will be in equilibrium with 200 units traded.

30. The diagram shows the market for CDs.



A government raises the minimum price of CDs from \$12 to \$14.

What will be the outcome?

- A Excess demand increases by 2 units.
- B Excess demand increases by 4 units.
- C Excess supply decreases by 2 units.
- D Excess supply increases by 2 units.

Question	Answer	Marks
1	With the help of a diagram, assess whether the benefits of a buffer stock scheme for an agricultural product always outweigh its disadvantages. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis Diagram of a buffer stock scheme, correctly labelled with an equilibrium (target) price and quantity and with changes to price and quantity when there is an increase and a decrease of supply. Explanation of the potential benefits of a buffer stock scheme. Explanation of the disadvantages and limitations associated with a buffer stock scheme and of why such a scheme might be ineffective. AO3 Evaluation • Consideration of the potential effectiveness of a buffer stock scheme. • Consideration of whether the benefits of a buffer stock scheme always outweigh its disadvantages. • In conclusion, whether such a scheme will always be effective for all agricultural products. Accept all valid responses.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
2	Assess the effectiveness of government intervention through buffer stock schemes in agricultural markets. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis (max 8 marks) The advantages and disadvantages of a buffer stock scheme in agricultural markets may include • Effective in smoothing out price changes, particularly where these are volatile • Responsive to market conditions • Involves minimum and maximum prices • Storage of products can be used to influence price changes • Can be inefficient when surplus stocks build up • Little incentive for producers to change what they produce • Could lead to waste of perishable products Other methods of intervention can include • Subsidies paid to agricultural producers • Tariffs on imports • Export controls Advantages and disadvantages include • Less intervention, more emphasis on market forces • Improved security of supply for domestic market • Reduced dependence on imports • Income of efficient farmers can increase • Subsidies involve an opportunity cost Level 2 max for 1 sided answers AO3 Evaluation (max 4 marks) • Consideration of relative effectiveness of buffer stock schemes compared to other methods of intervention • How to determine minimum prices for government intervention • Problem of over-supply and impact on future market prices • Informal markets and scale of subsistence producers can be issues in low-income countries Accept all valid responses. A one-sided response cannot gain any marks for evaluation.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
3	With the help of a diagram, assess whether the introduction of a minimum price in a market can be justified. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis Diagram of a minimum price in a market, correctly labelled with an equilibrium price and an equilibrium quantity and showing the existence of a minimum price above what would be the case with no government intervention. Explanation of the potential advantages and benefits of a minimum price, such as making the prices of demerit goods more expensive to discourage consumption and protecting the incomes of producers in that market. Explanation of the potential disadvantages and limitations of a minimum price, such as the creation of excess supply and the greater likelihood of producers becoming less efficient, and of why the introduction of a minimum price might be ineffective. Note: Omission of a diagram/inaccurate diagram can only achieve a maximum of Level 2. A one-sided response can only gain a maximum of Level 2. AO3 Evaluation • Evaluation of the potential effectiveness of the introduction of a minimum price in a market. • Evaluation may contain an assessment of alternative policies but may only be rewarded if the advantages and disadvantages of minimum pricing policy have been discussed. • In conclusion, whether the introduction of a minimum price in a market can be justified. A one-sided response cannot gain any marks for evaluation. Accept all valid responses.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
4	Governments sometimes fix maximum prices on goods for different reasons. Assess whether the use of a maximum price is always advantageous. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis The advantages of a maximum price might include: • The price of certain products, such as essential foods, can be limited by the establishment of a price ceiling. • The effect of this will be to make such items more affordable to people. • This will reduce the extent of poverty in a country. The disadvantages of a maximum price might include: • The impact of the maximum price will be to create a situation of excess demand in the market (this needs to be clearly explained). • This could lead to queuing or rationing. • It could lead to the creation of a black or illegal market. (Up to 8 marks) AO3 Evaluation • Consideration of the relative balance of advantages and disadvantages of the policy. • Consideration of the impact of this policy on producers and consumers. • In conclusion, consideration of whether the policy will be effective and always advantageous. Accept all valid responses. (Up to 4 marks)	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
5	A government wishes to keep the price of an essential food, such as rice, affordable to help low-income households. Assess whether a policy of fixing the maximum price for an essential food is likely to be more effective than a policy of making transfer payments to low-income households. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis (max 8 marks) The advantages and disadvantages of using a maximum pricing policy may include: • If the price is set below the equilibrium price and quantity, then it will make e.g., rice more affordable. • This will help low-income consumers and improve their living standards and improve their health BUT • It will create a shortage and informal markets may arise. • Governments may have to increase their spending to enforce the legislation. • Rich and poor households gain equally. Transfer payments may include: • Unemployment benefits • Pensions • Food vouchers etc. The advantages and disadvantages include: • They can be targeted at those most in need. • They can be targeted at the purchase of basic foodstuffs BUT • May be costly and have an opportunity cost. • The policies may not be used for their intended purposes (even vouchers may be sold in illegal markets) A one-sided response can only gain a maximum of Level 2. AO3 Evaluation (max 4 marks) • Consideration of the relative effectiveness of a maximum pricing policy compared with the use of transfer payments. • Consider the impact on low-income consumers of basic foodstuffs • In conclusion, which policy is most likely to be effective in helping low-income consumers. Accept all valid responses. A one-sided response cannot gain any marks for evaluation.	12

SECTION B

1.	C	11.	B	21.	D
2.	D	12.	C	22.	D
3.	D	13.	C	23.	D
4.	B	14.	D	24.	D
5.	C	15.	A	25.	A
6.	A	16.	D	26.	A
7.	B	17.	A	27.	B
8.	C	18.	D	28.	B
9.	B	19.	C	29.	A
10.	C	20.	D	30.	D